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Editors-in-Chief
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Dear Editors-in-Chief,

I am pleased to submit the manuscript **“Post-Execution Defect Attribution: An Empirical Comparison of Four Methods on 6,000 Synthetic Failure Events”** for consideration as a Regular Article in *Empirical Software Engineering*.

Contribution. The paper presents an empirical comparison of four post-execution defect-attribution methods: pre-execution prediction score, exception-type heuristic, diff-based attribution, and a fused calibrated model. On 6,000 synthetic failure events derived from real defect labels (296,457 file instances mined from five mature open-source repositories), the fused model achieves an Expected Calibration Error (ECE) of 0.0157 versus 0.0936 for the pre-execution baseline—a six-fold improvement in calibration. The manuscript includes multi-seed sensitivity analysis, feature ablation, reliability diagrams, per-repository precision breakdowns, an industrial deployment walk-through with 12-month milestones, an operational failure-modes taxonomy (five distinct modes), a practitioner decision framework (five questions), comparison against manual triage workflows, long-term sustainability and maintenance-cost analysis, and an ethical-considerations section.

Fit for EMSE. The contribution is a rigorous empirical software-engineering study with multi-method comparison, full reproducibility artefacts, an explicit threats-to-validity treatment, and direct engagement with the practical deployment concerns the EMSE community emphasises. The methodology connects defect prediction (a heavily studied SE problem) with the practitioner-relevant task of triaging test failures under uncertainty, and reports calibration metrics (ECE, reliability diagrams) that the empirical SE community has been increasingly emphasising over raw accuracy.

Reproducibility. The underlying defect-prediction dataset is permanently archived on Zenodo with DOI [10.5281/zenodo.19682732](https://doi.org/10.5281/zenodo.19682732). The post-execution attribution code, synthetic failure-event generator, calibration analysis, and reliability-diagram scripts are public at <https://github.com/javvadivijayprasad/DefectAnalysisReserch>.

Honest disclosure. The failure events are synthesised from real defect labels rather than mined from real CI history; this is stated in the Threats to Validity section. The trained ML model is the production model deployed in our industrial test-prioritisation platform, with defect labels derived from commit-message keyword matching rather than linked issue trackers. The mining of real CI failure logs at comparable scale is explicitly framed as future work.

Originality and exclusivity. This manuscript has not been published previously and is not

under consideration at any other journal or conference. A companion paper on test prioritisation is in camera-ready preparation at *IEEE Software*; a separate empirical evaluation of the underlying classifier is under review at *IEEE Access*; a cross-repository transferability study is under preparation for *Software Testing, Verification & Reliability* (Wiley). These companion papers are referenced for context; the post-execution attribution contribution is unique to this manuscript.

Use of AI tools. In preparation of this manuscript, I used Anthropic’s Claude (a large language model) for drafting assistance and copy-editing of the prose. All empirical claims, data, and statistical analyses were generated by my own scripts and verified against the on-disk artefacts; the AI tool was not used to generate data or results. This is disclosed in the Acknowledgments section of the manuscript.

Conflicts of interest. None.

Funding. No external funding was received.

Suggested associate editor topics. Empirical software engineering, defect prediction, test triage, calibration, mining software repositories.

I would be grateful for your editorial consideration and look forward to the reviewers’ feedback. Thank you for your time.

Sincerely,

Vijay Prasad Javvadi